VITALIS Panel Drawing Page Template (v4 — explicit layout)

Purpose: Define the universe of pages the generator may emit, the conditions under which each page is included, and where in the drawing set it sits.

Baseline: Five drawing sets (YCH + 4 newer jobs) + Randall's clarifications + explicit layout rules from this round.

Reference drawing sets

Set	Project	Brand / HMI	Pages	Power	Notable options observed
E1399 YCH rev E	Freecovery / YCH	Danfoss / AK-SM850	85	460 V, 75 HP	Both Total Failure pages
E1395 LAZY ACRES rev J	LAZY ACRES (MB0219)	Danfoss / AK-SM850	76	208 V, 60 HP	Smallest set
E1380 BRIGHT RENEWABLES COOLSHIFT rev G	ROXSTA G6	Danfoss / **MMIGRS2**	86	575 V, 75 HP	Energy meter (BOM), 2× EKE-1C
E1377 VALLARTA 75 rev 2 (`ADDON_BOARD1`)	ROXSTA G6	**Copeland / E3**	82	460 V, 75 HP	UPS, CO2 leak, Spare 810-3064
E1392 4S RANCH rev J1	4S RANCH (MB0216)	**Copeland / E3**	81	208 V, 30 HP	UPS, CO2 leak, Energy meter w/ Rogowski

Source of truth for option inputs: `DWG_AUTOMATION.xlsx`.

> **Note on page-count variance.** Per-set variance across these five sets is partly options + slot count, and partly **human engineering choices and reference errors** during drafting. Treat structural inconsistencies between sets as drafting variation, not as evidence of conditional logic, unless the variation correlates with a known option.

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1. Layout rules (the explicit page-composition contract)

These are the firm rules the generator follows for page composition. They override what appears in any particular existing set (where drafters compressed multiple things onto one page for layout reasons).

Rule 1 — 3-phase distribution: two compressors per page, one top + one bottom

Each 3-phase distribution page carries exactly two compressor circuits — one in the upper half of the page, one in the lower half. This rule applies regardless of VFD vs across-the-line topology.

Slot reservation: to support up to 6 MT + 6 LT = **12 compressor slots**, the generator reserves **6 three-phase distribution pages** total (pairs: MT1+MT2, MT3+MT4, MT5+MT6, LT1+LT2, LT3+LT4, LT5+LT6). The pages always reserve both upper and lower halves. If a compressor in a slot is disabled (`QTY/Enable == FALSE`), its half of the 3φ page is drawn as INTENTIONALLY LEFT BLANK rather than collapsed.

Pairing convention: odd-numbered compressor (MT1, MT3, MT5, LT1, LT3, LT5) goes in the top half; even-numbered (MT2, MT4, MT6, LT2, LT4, LT6) in the bottom half. This is deterministic — the slot-to-page-half mapping does not vary by job.

Rule 2 — "Other 3φ items" page

After the 6 compressor-pair 3φ pages, **one dedicated page** for the other 3-phase items. Content lands here:

- **Grid Monitoring Relay** (CR??51 + GMR + FU??51 fuse) — currently jammed onto the bottom of MT Comp 1's 3φ page in YCH; moves to this dedicated page going forward
- **Control transformers** (T?031 3kVA 120VAC + T?071 500VA 24VAC) — currently on the last 3φ page in YCH; move here
- **Energy Meter** (PM5101 EEM-MB371-24DC + 3× Rogowski coils on V1/V2/V3 + CR??141 ROGOWSKI FAULT relay + RJ45 X1-ETH) if `EnergyMeter == TRUE`

This is the page that contains everything 3-phase-electrical that isn't a compressor.

Rule 3 — 120VAC items page

After "other 3φ items", **one page** dedicated to 120VAC items. Content lands here:

- WLT* enclosure light
- MTR* + TAS* enclosure fan + thermostat
- RECP* receptacle (15A)

- ES?291 Phoenix FL Switch 2000 (Ethernet switch) if `EthernetSwitch == TRUE`
- 24VDC PSUs (brand-determined per §3):
    - Danfoss: PWS?101 + PWS?151 (AK-PS250 × 2) + PWS?221 (WAGO 2587-2144)
    - Copeland: PWS?221 (WAGO 2587-2144) only
- E-stop circuit (External + Local Machine Shutdown chain with CR??41 MACHINE SHUTDOWN RELAY; XB5 button replaces JUMPER if `MachineStopButton == TRUE`)
- Safety relay (CR??81)

Anything UPS-protected lives on the next page.

Rule 4 — UPS page

If `UPS == TRUE`, **one page** after 120VAC items: `120VAC UPS TO TRANSFORMER SUPPLIES`. Contents per §5.4.

If `UPS == FALSE`, this page slot does not exist (no INTENTIONALLY LEFT BLANK — the page is fully absent).

Rule 5 — All other drawings follow

The rest of the page sequence (rack-general pages, compressor bundles, oil mgmt, HR, gas cooler, regulation, controller block, alarms, BOM, layout) continues per §5 below.

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2. Slot reservations — 6 MT + 6 LT going forward

The previous template assumed up to 7 MT + 6 LT based on the current spreadsheet. Going forward, **reserve 6 MT + 6 LT slots** in the generator. This adjusts the spreadsheet (drop MT7) and fixes the slot count at a round 12.

Rack	Slots reserved	Per-slot bundle size	Total bundle pages reserved
MT	6	4 sheets	24
LT	6	4 sheets	24

Plus the 3φ pages: 6 compressor pairs = 6 pages (covers MT1+2, MT3+4, MT5+6, LT1+2, LT3+4, LT5+6).

For a fully-populated 12-compressor panel, the per-compressor footprint of the drawing set is:

- 6 pages of 3-phase distribution (compressor pairs)
- 48 pages of compressor bundles (12 × 4)

That's 54 reserved compressor-related slots. For smaller panels, the same 54 slots are reserved but unused compressors get INTENTIONALLY LEFT BLANK pages.

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3. Brand split — controller block architecture

Two architectures by `Brand`:

- **Danfoss** (YCH, E1395, E1380): 9-sheet controller block — AK-PC-782B main + 6× AK-XM modules + AK-CM-101C COM + HMI (AK-SM850 or MMIGRS2). EKE-1C stepper drivers are an **independent** option (not coupled to HMI choice).
- **Copeland** (E1377, E1392): 5–7-sheet block — E3 PIB + MultiFlex 168A0 + (optional) MultiFlex 168 (spare board) + IPRO CO2 HP Controller + IPRO XEV20D × 2 + (optional) CO2 leak page.

The 24VDC PSU set is also brand-determined: Danfoss = 2× AK-PS250 + WAGO; Copeland = WAGO only.

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4. Option inventory

4.1 Per-compressor (sheet: `Project Info`)

Column	Effect when TRUE	Effect when FALSE
`VFD`	3φ circuit uses VFD template + 7-terminal signal strip	Across-the-line template
`Door Lights`	Emit W/R lamps on Control sheet + door layout slot	Omit lamps + slot
`Door Off/Auto Switch`	Emit TG selector on Control sheet + door layout slot	Omit selector + slot
`QTY/Enable`	Emit 4-sheet bundle + emit top/bottom of paired 3φ page normally	Emit 4× BLANK pages + 3φ page half BLANK

`FLA`	Drives CB / contactor / VFD sizing + motor block FLA attribute	n/a

> Door Lights and Door Off/Auto Switch are **panel-wide in practice** — Randall has never shipped a panel where these differ between compressors on the same job. The per-compressor representation is kept for future-proofing.

4.2 Existing panel-level options (`Panel Options`)

Option	Where in drawing set
UPS (Phoenix QUINT4)	Dedicated UPS page after 120VAC items page (Rule 4)
6" Legs / 18" D enclosure	Layout page only
Energy Meter (EEM-MB371-24DC)	"Other 3φ items" page (Rule 2)
Ethernet Switch (Phoenix FL Switch 2000)	120VAC items page (Rule 3)
Machine Stop Button (Harmony XB5)	120VAC items page (Rule 3) — replaces JUMPER in safety chain
Spare 16I/8RO Board *(renamed from "8I/8O")*	**Copeland:** MF73061 page in controller block. **Danfoss:** additional AK-XM-205A. Max 1 per panel
CO2 Leak Detection	**Copeland:** dedicated OPTIONAL CO2 SENSOR page (renamed from "FIELD INSTALLED") with MRLDS-450. **Danfoss:** ❓ no example panel yet

4.3 New options to add to the spreadsheet

New column	Default	Purpose
`MTTotalFailure`	TRUE	Emit MT Total Failure Rack / Fast Return page
`LTTotalFailure`	TRUE	Emit LT Total Failure Rack / Fast Return page
`HeatRecovery`	TRUE	Emit the 4-page HR block (EXTERNAL REQUEST, HR1 SENSORS, HR1 3-WAY VALVE, WATER PUMP)
`LiquidSubcoolerInjection`	FALSE	Emit Liquid Subcooler Injection page (Copeland only; Danfoss has no slot)

`EKE1C_Steppers`	FALSE	Emit N1-EKE-1C + N2-EKE-1C pages (Danfoss only)
`CO2LeakDetection`	FALSE	Convert from current free-text "Special Requirements/Notes" to structured boolean

4.4 Items now standard (no flag needed)

- **GROUP 2 Gas Cooler Fan Control** — emit unconditionally on every panel (Randall: now standard)

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5. Invariant page sequence (final, with layout rules applied)

The full page sequence the generator emits, top-down. Items marked 🆕 are new spreadsheet columns from §4.3.

5.1 Front matter

Slot	Page	Gate
0	REVISION TRACKING	always
1	COVER SHEET	always

5.2 3-phase power distribution — six paired-compressor pages (always)

Per **Rule 1** (Layout): two compressors per page, odd in upper half, even in lower half.

Slot	Page	Top half	Bottom half
2	3-PHASE POWER DISTRIBUTION	MT1 (slot enabled or BLANK)	MT2
3	3-PHASE POWER DISTRIBUTION	MT3	MT4
4	3-PHASE POWER DISTRIBUTION	MT5	MT6
5	3-PHASE POWER DISTRIBUTION	LT1	LT2
6	3-PHASE POWER DISTRIBUTION	LT3	LT4
7	3-PHASE POWER DISTRIBUTION	LT5	LT6

If a compressor slot is disabled (`QTY/Enable[n] == FALSE`), that page half is INTENTIONALLY LEFT BLANK. **If both halves of a page are disabled, the page is still emitted with both halves BLANK** (consistent with the slot-reservation pattern).

Each compressor circuit (when populated) follows one of two templates:

- **VFD template:** CB → contactor → VFD → motor + 7-terminal signal strip (B?-X? positions 1-6, PE)
- **Across-the-line template:** CB → contactor → motor (no terminal strip)

5.3 "Other 3φ items" page (always — per Rule 2)

Slot	Page	Gate
8	3-PHASE POWER DISTRIBUTION — OTHER (suggested title)	always

Content:

- **Main disconnect breaker** + incoming L1/L2/L3 from grid
- **PDB-A / PDB-B** power distribution blocks
- **Grid Monitoring Relay** (CR??51 + GMR + FU??51)
- **Control transformers:** T?031 (3kVA 120VAC primary) + T?071 (500VA 24VAC primary) with primary fuses on PDB-B
- **Energy Meter** if `EnergyMeter == TRUE`: PM5101 EEM-MB371-24DC + RC?101/RC?121/RC?141 Rogowski coils on V1/V2/V3 + CR??141 ROGOWSKI FAULT relay + RJ45 X1-ETH to ethernet switch

5.4 120VAC items page (always — per Rule 3)

Slot	Page	Gate
9	120VAC DISTRIBUTION	always

Content (in approximate top-to-bottom order on the page):

- WLT?021 enclosure light + CB?021
- MTR?041 enclosure fan + TAS?041 thermostat + CB?041
- RECP?061 15A receptacle + CB?061
- ES?291 Phoenix FL Switch 2000 if `EthernetSwitch == TRUE`
- 24VDC PSUs:
    - if `Brand == Danfoss`: PWS?101 + PWS?151 (AK-PS250 × 2, both with 6A breakers) + PWS?221 (WAGO 2587-2144)
    - if `Brand == Copeland`: PWS?221 (WAGO 2587-2144) only
- EXTERNAL MACHINE SHUTDOWN — JUMPER — LOCAL MACHINE SHUTDOWN — CR??41 MACHINE SHUTDOWN RELAY safety chain:

    - if `MachineStopButton == TRUE`: XB5 button in place of JUMPER
    - if `MachineStopButton == FALSE`: JUMPER installed
- CR??81 safety relay

5.5 UPS page (UPS option — per Rule 4)

Slot	Page	Gate
10 (if UPS on)	120VAC UPS TO TRANSFORMER SUPPLIES	`UPS == TRUE`

Content:

- UPS70?1 1kVA single-phase 120V→120V
- 2× external 24VDC battery modules (BAT?031, BAT?071)
- CR?071 (120VAC ALARM) + CR?072 (BAT. MODE) dry contacts to controller DIs
- 5× 24VAC 50VA transformers (T?111, T?131, T?151, T?171, T?191) feeding 24VAC2–24VAC6 rails (Copeland uses these heavily; Danfoss may use fewer downstream rails — same page structure regardless)
- Primary CBs and secondary 24VAC breakers per transformer

If `UPS == FALSE`, this slot is absent — page sequence renumbers everything after it.

5.6 MT general pages (if any MT slot enabled)

Page	Gate
SUCTION GAS MONITORING, LOW PRESSURE LIMITER (B0)	any MT enabled
DISCHARGE GAS MONITORING (B0)	any MT enabled
POST INJECTION (B0)	`MTPostInjection == TRUE` (future spreadsheet column; default TRUE)
TOTAL FAILURE RACK / FAST RETURN (B0)	`MTTotalFailure == TRUE` 🆕

5.7 MT compressor bundles (always emit 4 sheets per slot × 6 slots)

For each MT slot 1..6:

- if `QTY/Enable[MTn] == TRUE`: emit 4 sheets normally
- if `QTY/Enable[MTn] == FALSE`: emit 4× INTENTIONALLY LEFT BLANK

Bundle:

1. `COMPRESSOR n CRANKCASE HEATER / CAPACITY CONTROL`
2. `SAFETY CHAIN AND CONTROL` (HP limiter + motor protection — SE-B1/2 for MT)
3. `SAFETY CHAIN AND CONTROL` (Traxoil + Delta PII; jumper note if no Delta PII)
4. `CONTROL` (run/fail pilots + I-O-II selector — gated by `Door Lights` and `Door Off/Auto Switch`)

5.8 LT general pages (if any LT slot enabled)

Page	Gate
SUCTION GAS MONITORING, LOW PRESSURE MONITOR (G0)	any LT enabled
DISCHARGE GAS MONITORING (G0)	any LT enabled
TOTAL FAILURE RACK / FAST RETURN (G0)	`LTTotalFailure == TRUE` 🆕

5.9 LT compressor bundles (always emit 4 sheets per slot × 6 slots)

Same as §5.7 for LT slots 1..6. Architectural difference: INT69 motor protection (not SE-B1/2), no Delta PII, 47/34 bar HP cutout.

5.10 Oil management + HR + gas cooler + regulation

Page	Gate
OIL MANAGEMENT	always
OPTIONAL OIL MANAGEMENT	content gated by future `OilSepHeater` column; page slot always emitted
EXTERNAL REQUEST	`HeatRecovery == TRUE` 🆕
HR1 SENSORS	`HeatRecovery == TRUE` 🆕
HR1 3-WAY VALVE	`HeatRecovery == TRUE` 🆕
WATER PUMP	`HeatRecovery == TRUE` 🆕
GROUP 2 CONTROL (gas cooler)	always (now standard)

GROUP 1 CONTROL (gas cooler)	always
BYPASS VALVE & SENSORS (P1, gas cooler)	always — NTC10K if Copeland, PT1000 if Danfoss; bypass-valve content gated by future column, sensors always present
PRESSURE TRANSDUCER (HP, R1)	always
VALVE 1 (HP, R1)	always — Danfoss CCMT16 / Copeland IPRO XEV20D channel
VALVE 2 (HP, R1)	always
PRESSURE TRANSDUCER (MP, S1)	always
VALVE 1 (MP, S1)	always — Danfoss CCM40 / Copeland XEV20D
VALVE 2 (MP, S1)	always

5.11 Refrigerant collector + subcooler

Page	Gate
SENSORS (refrigerant collector S2)	always
LIQUID SUBCOOLER INJECTION (S2)	`LiquidSubcoolerInjection == TRUE` 🆕. **Copeland reserves the slot when OFF** (emit BLANK); Danfoss does not reserve the slot

5.12 Controller block — Danfoss

If `Brand == Danfoss`:

1. AK-PC-782B (Module Address 1)
2. AK-XM-103A addr 2
3. AK-XM-103A addr 3
4. AK-XM-101A addr 4
5. AK-XM-205A addr 5
6. AK-XM-208C addr 6
7. AK-CM-101C COM
8. AK-XM-205A addr 7

9. AK-XM-205A addr 8 if `SpareIOBoard == TRUE` 🔶 (insertion order inferred)
10. If `EKE1C_Steppers == TRUE` 🆕: N1-EKE-1C + N2-EKE-1C (2 pages)
11. HMI:
    - `Interface == AK-SM850` → AK-SM850 page
    - `Interface == MMIGRS2` → MMIGRS2 page

Module catalog: AK-PC-782B (fixed), AK-XM-103A (4 AI + 4 AO), AK-XM-101A (8 AI), AK-XM-205A (4 AI + 8 DO), AK-XM-208C (4 AI + 4 SO), AK-CM-101C (comms).

5.13 Controller block — Copeland

If `Brand == Copeland`:

1. E3 POWER INTERFACE BOARD (E3 controller, COM 1–4 RS-485, ethernet, IO?081 plugin I/O card)
2. MF68061 MultiFlex 168A0 (Copeland 810-3065 — 16I/8RO/4AO, standard)
3. MF73061 MultiFlex 168 (Copeland 810-3064 — 16I/8RO) if `SpareIOBoard == TRUE`
4. IPRO CO2 HIGH PRESSURE CONTROLLER (IP70031, 818-9008) + Visograph display VG5714170201
5. IPRO XEV20D DUAL STEPPER CONTROLLER 1 (HP valves via CAN)
6. IPRO XEV20D DUAL STEPPER CONTROLLER 2 (MP valves via CAN)
7. OPTIONAL CO2 SENSOR (renamed from FIELD INSTALLED) if `CO2LeakDetection == TRUE` — MRLDS-450 wiring

5.14 Alarms (always)

Page	Gate
ALARMS / MTR ROOM	always — exhaust fan relay, alarm light (R), alarm bell, motor room leak light (B)

5.15 BOM + layout (always)

Page	Gate
BILL OF MATERIAL	always
ENCLOSURE LAYOUT	always (6" legs detail when `Legs18D == TRUE`)
DOOR LAYOUT	always (HMI cutout per `Interface`, machine stop slot per `MachineStopButton`, per-compressor lamp/selector slots)

DOOR DETAILS 1 / DOOR LAYOUT ZOOM	always
LAYOUT DETAILS 2 / CIRCUIT PROTECTION DEVICES	always
LAYOUT DETAILS 3 / CONTROL RELAYS	always
DANFOSS HARDWARE / DANFOSS CONTROL DEVICES	Danfoss only
MOTOR PROTECTION DEVICES / COMPRESSOR MOTOR COMBO STARTERS	always
CONTACTOR RELAYS	Copeland only (naming)

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6. Generator algorithm (final)

```None
emit_revision_tracking()                                          #
slot 0
emit_cover_sheet(electrical_specs, contractor, brand, hmi)        #
slot 1

# 3φ distribution: 6 reserved compressor-pair pages
# Slot map: page_2=(MT1,MT2), page_3=(MT3,MT4), page_4=(MT5,MT6),
#           page_5=(LT1,LT2), page_6=(LT3,LT4), page_7=(LT5,LT6)
for pair in
[(MT,1,2),(MT,3,4),(MT,5,6),(LT,1,2),(LT,3,4),(LT,5,6)]:
    emit_3phase_pair_page(
        top=compressor_slot(pair.rack, pair.odd),       # enabled
or BLANK
        bottom=compressor_slot(pair.rack, pair.even)    # enabled
or BLANK
    )

# "Other 3φ items" — slot 8
emit_3phase_other_page(
```

```
    grid_monitoring=True,                               # always
    control_transformers=True,                          # T?031 +
T?071, always
    energy_meter=panel_options.energy_meter,            # PM5101 +
Rogowski + RJ45 if true
    pdb_a_b=True
)

# 120VAC items — slot 9
emit_120vac_items_page(
    machine_stop_button=panel_options.machine_stop,     # JUMPER vs
XB5
    ethernet_switch=panel_options.ethernet_switch,
    psu_set=(
        [PWS_AK_PS250, PWS_AK_PS250, PWS_WAGO_2587_2144] if brand
== Danfoss
        else [PWS_WAGO_2587_2144]
    )
)

# UPS — slot 10 if option on
if panel_options.ups:
    emit_ups_supply_page(brand=brand, battery_modules=2)

# MT side
if any(compressors_enabled.mt):
    emit_mt_suction_gas()
    emit_mt_discharge_gas()
    if panel_options.mt_post_injection: emit_post_injection()
    if panel_options.mt_total_failure: emit_mt_total_failure()   #
🆕
for mt_slot in 1..6:
    if compressors_enabled.mt[mt_slot]:
        emit_compressor_bundle(rack=MT, slot=mt_slot, ...)
    else:
        emit_n_blank_pages(4)
```

```
# LT side
if any(compressors_enabled.lt):
    emit_lt_suction_gas()
    emit_lt_discharge_gas()
    if panel_options.lt_total_failure: emit_lt_total_failure()   # 
🆕
for lt_slot in 1..6:
    if compressors_enabled.lt[lt_slot]:
        emit_compressor_bundle(rack=LT, slot=lt_slot, ...)
    else:
        emit_n_blank_pages(4)

# Oil management + HR + gas cooler + HP/MP regulation
emit_oil_management()
emit_optional_oil_management()
if panel_options.heat_recovery:                                  # 
🆕
    emit_external_request()
    emit_hr1_sensors()
    emit_hr1_3way_valve()
    emit_water_pump()
emit_gas_cooler_group2()                                         # 
now standard
emit_gas_cooler_group1()
emit_bypass_valve_and_sensors(
    sensor_type=(NTC10K if brand == Copeland else PT1000),
    include_bypass_valve=panel_options.gas_cooler_bypass_valve   # 
future column
)
emit_hp_regulation(brand=brand)
emit_mp_regulation(brand=brand)
emit_refrigerant_collector_sensors()

if brand == Copeland:
```

```
    if panel_options.liquid_subcooler_injection:                    #
🆕
        emit_liquid_subcooler_page()
    else:
        emit_liquid_subcooler_blank_slot()                          #
Copeland reserves slot

# Brand-keyed controller block
if brand == Danfoss:
    emit_danfoss_controller_block(
        compressors_enabled,
        options=panel_options,
        hmi=hmi,                                                    #
AK-SM850 or MMIGRS2
        eke1c_steppers=panel_options.eke1c,                         #
🆕 independent of HMI
        spare_io_board=panel_options.spare_io_board
    )
elif brand == Copeland:
    emit_copeland_controller_block(
        compressors_enabled,
        options=panel_options,
        spare_addon_board=panel_options.spare_io_board,
        co2_leak_detection=panel_options.co2_leak_detection      #
emits OPTIONAL CO2 SENSOR
    )

emit_alarms()
emit_bom()
emit_enclosure_layout(legs=panel_options.legs_18d)
emit_door_layout(
    hmi=hmi,
    compressors_with_door_lights,                                   #
panel-wide in practice
    compressors_with_off_auto_switch,                               #
panel-wide in practice
```

```
    machine_stop=panel_options.machine_stop
)
emit_layout_details(brand=brand)
```

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7. Spreadsheet changes required

New columns on `Panel Options`

Column	Default	Notes
`MTTotalFailure`	TRUE	
`LTTotalFailure`	TRUE	
`HeatRecovery`	TRUE	
`LiquidSubcoolerInjection`	FALSE	Copeland-only behavior
`EKE1C_Steppers`	FALSE	Danfoss-only
`CO2LeakDetection`	FALSE	Convert from current free-text Special Requirements/Notes

Renames

Existing	New
"Spare 8 Input / 8 Output Board" (Copeland half)	"Spare 16 Input / 8 Relay Output Board (MF73061)"
(page title "FIELD INSTALLED" in E1392 template)	"OPTIONAL CO2 SENSOR"

Slot-count change

Existing	New
7 MT + 6 LT compressor slots in `Project Info`	**6 MT + 6 LT** (drop MT7 row)

Removed from options (now standard)

- `GROUP 2 CONTROL` gas cooler fan page — emit unconditionally

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8. Page-slot map (high-level)

For a fully-populated 12-compressor Danfoss panel with all options enabled, the slot map is approximately:

Slot	Content
0	Revision Tracking
1	Cover Sheet
2–7	3-phase distribution (6 compressor-pair pages)
8	3-phase distribution — other (grid mon, xfmrs, energy meter)
9	120VAC distribution
10	120VAC UPS to transformer supplies
11–14	MT general (suction, discharge, post inj, total failure)
15–38	MT compressor bundles (6 slots × 4 sheets = 24 sheets)
39–41	LT general (suction, discharge, total failure)
42–65	LT compressor bundles (6 slots × 4 sheets = 24 sheets)
66	Oil Management
67	Optional Oil Management
68–71	HR1 block (external request, sensors, 3-way valve, water pump)
72	Gas Cooler Group 2
73	Gas Cooler Group 1
74	Bypass Valve & Sensors
75–77	HP regulation (transducer + 2 valves)

78–80	MP regulation (transducer + 2 valves)
81	Refrigerant Collector Sensors
82	Liquid Subcooler Injection (Copeland — would be in different position on Copeland panels)
83–91	Controller block (Danfoss: 9 sheets — PC-782B, 6× XM, CM, HMI)
(+2 if EKE-1C)	EKE-1C pages
92	Alarms
93	Bill of Material
94	Enclosure Layout
95	Door Layout
96–100	Layout details + motor protection details

This is the **reservation skeleton**. Disabled options pop their pages; enabled options that aren't in this list (e.g. spare IO board, CO2 sensor on Copeland) push subsequent pages down.

For a fully-empty (no compressors enabled, all options off) panel, the same skeleton applies — most pages emit as `INTENTIONALLY LEFT BLANK`. This matches the reserved-slot behavior the existing drawing sets demonstrate.

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9. Items still genuinely unknown

1. **Danfoss with spare board** — AK-XM-205A addr 8 insertion order inferred, not observed.
2. **Bypass valve content when ON** — only seen in OFF state (sensors-only).
3. **Danfoss-path CO2 leak detection** — no Danfoss panel has it; wiring TBD.
4. **6" legs layout detail** — not in any of the five reference sets.
5. **Drafting variations** — sets show minor reference errors (compressor 4 mislabeled as 1, etc.); treat as drafter artifact, not generator behavior.

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10. Provenance

Pages directly rasterized for layout verification:

Set	Sheet	Page title	Confirms
E1377	7	120VAC UPS TO TRANSFORMER SUPPLIES	UPS schematic structure
E1377	57	BYPASS VALVE & SENSORS	NTC10K sensors (Copeland)
E1377	66	LIQUID SUBCOOLER INJECTION	BLANK reserved slot
E1377	67	E3 POWER INTERFACE BOARD	Copeland main controller
E1377	70	810-3064 BOARD	MF73061 spare board layout
E1377	71	MULTIFLEX 168	MRLDS-450 CO2 sensor wiring
E1377	72	IPRO CO2 HIGH PRESSURE CONTROLLER	IP70031
E1377	73	IPRO XEV20D DUAL STEPPER 1	CAN bus valve actuator
E1380	6	120VAC DISTRIBUTION	Machine shutdown JUMPER, Danfoss dual PSU
E1392	5	3-PHASE POWER DISTRIBUTION (last)	PM5101 + Rogowski + RJ45 ethernet
E1395	67	AK-SM850	Danfoss System Manager
E1395	68	ALARMS / MTR ROOM	Standard alarms

Cross-set page-title comparison: `page-title-comparison.txt` (separate file).

Layout rules introduced this round (v3 → v4):

- Two compressors per 3φ page (odd top, even bottom)
- 6 MT + 6 LT slot reservations going forward

- "Other 3φ items" as a dedicated page (grid monitor, xfmrs, energy meter)
- "120VAC items" as a dedicated page (lights, receptacle, PSUs, machine stop, ethernet switch)
- UPS as a dedicated next page when option is on
- Pages 2–10 are now structurally fixed regardless of panel content